

Institute Name :

Smt. Chandaben Mohanbhai Institute of Computer Applications

Department Name :

MCA

Subject :

CAUC306- Enterprise Computing with Java EE

Semester :

1

Assignemnt Name :

CLASSES AND OBJECT

Starting Date :

20-07-2026

Submission Date :

14 – 08 – 2026

Student ID :

26MCA095

Student Name :

Pajvani Krisha Hirenbhai

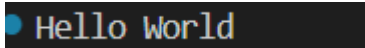
Part 1: Structure of Java Program

Exercise 1: Hello World Program

Write a Java program to display "Hello, World!" on the console.

>>

```
public class Hello{  
    public static void main (String[] args){  
        System.out.println("Hello World");  
    }  
}
```

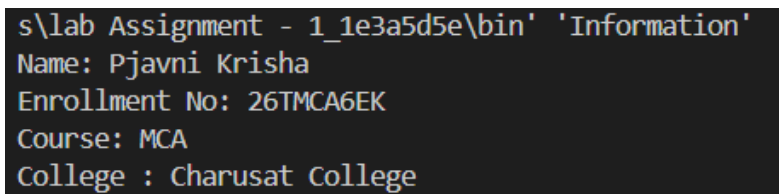


Exercise 2: Display Personal Information

Write a Java program to display your name, Enrolment number, course, and college name.

>>

```
public class Information {  
    public static void main(String[] args){  
        System.out.println("Name: Pjavni Krisha");  
        System.out.println("Enrollment No: 26TMCA6EK");  
        System.out.println("Course: MCA");  
        System.out.println("College : Charusat College");  
    }  
}
```

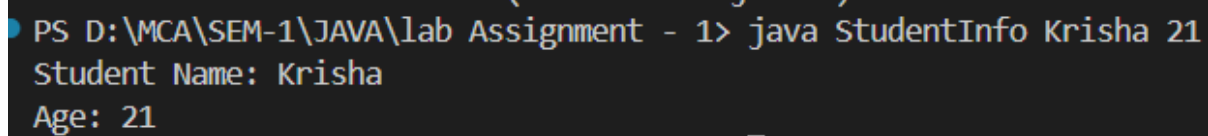


Exercise 3: Understanding Command Line Arguments

Write a Java program that accepts a student's name and age as command-line arguments and displays them.

>>

```
public class StudentInfo {  
    public static void main(String[] args) {  
        String name = args[0];  
        int age = Integer.parseInt(args[1]);  
  
        System.out.println("Student Name: " + name);  
        System.out.println("Age: " + age);  
    }  
}
```



The screenshot shows a terminal window with a dark background. The command prompt is 'PS D:\MCA\SEM-1\JAVA\lab Assignment - 1>'. The command entered is 'java StudentInfo Krishna 21'. The output of the program is displayed on two lines: 'Student Name: Krishna' and 'Age: 21'.

Exercise 4: Arithmetic Operations

Write a Java program to perform addition, subtraction, multiplication, division, and modulus operations on two numbers entered by the user.

>>

```
public class Arithmetic {  
    public static void main(String[] args) {  
        int a = Integer.parseInt(args[0]);  
        int b = Integer.parseInt(args[1]);  
  
        int sum = a + b;  
        int difference = a - b;  
        int product = a * b;  
        int division = a / b;  
        int modulus = a % b;  
  
        System.out.println("Sum: " + sum);  
        System.out.println("Difference: " + difference);  
        System.out.println("Product: " + product);  
        System.out.println("Division: " + division);  
        System.out.println("Modulus: " + modulus);  
    }  
}
```

```
}}
```

```
● PS D:\MCA\SEM-1\JAVA\lab Assignment - 1> java Arithmetic 5 10
Sum: 15
Difference: -5
Product: 50
Division: 0
Modulus: 5
```

Exercise 5: Demonstrating Java Program Structure

Write a Java program illustrating the use of package declaration, import statement, class declaration, main() method, comments, and output statements.

```
>>
```

```
import java.util.Scanner;

public class StructureDemo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Java Program Structure");

        System.out.print("Enter your name: ");
        String name = sc.nextLine();

        System.out.println("Hello, " + name);

        sc.close();
    }
}
```

```
● Java Program Structure
Enter your name: krisha
Hello, krisha
```

Exercise 6: Using Scanner Class

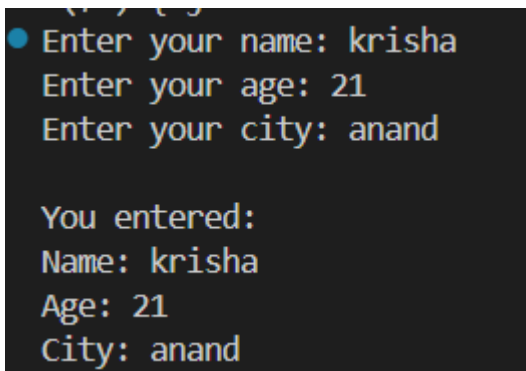
Write a Java program that accepts user input using the Scanner class and displays the entered values.

```
>>
```

```
import java.util.Scanner;

public class StructureDemo {
```

```
public static void main(String[] args) {  
  
    Scanner sc = new Scanner(System.in);  
  
    System.out.print("Enter your name: ");  
    String name = sc.nextLine();  
  
    System.out.print("Enter your age: ");  
    int age = sc.nextInt();  
  
    System.out.print("Enter your city: ");  
    sc.nextLine();  
    String city = sc.nextLine();  
  
    System.out.println("\nYou entered:");  
    System.out.println("Name: " + name);  
    System.out.println("Age: " + age);  
    System.out.println("City: " + city);  
    sc.close();  
}  
}
```



A screenshot of a terminal window with a dark background and light-colored text. It shows the execution of a Java program. The first three lines are prompts for user input: "Enter your name:", "Enter your age:", and "Enter your city:". The user has entered "krisha", "21", and "anand" respectively. The next three lines are the program's output: "You entered:", "Name: krisha", "Age: 21", and "City: anand".

Exercise 7: Type Conversion

Write a Java program demonstrating implicit and explicit type casting.

>>

```
public class TypeConversion {  
    public static void main(String[] args)  
    {  
        int a = 10;
```

```

double b = a;

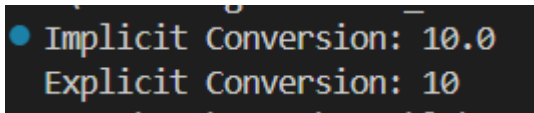
double x = 10.5;

int y = (int) x;

System.out.println("Implicit Conversion: " + b);

System.out.println("Explicit Conversion: " + y);
}
}

```



```

● Implicit Conversion: 10.0
  Explicit Conversion: 10

```

Exercise 8: Mathematical Functions

Write a Java program to demonstrate the use of methods from the Math class such as sqrt(), pow(), max(), min(), and random().

>>

```

public class MathematicaFunction {

    public static void main(String[] args) {

        System.out.println(Math.sqrt(16));

        System.out.println(Math.pow(2, 3));

        System.out.println(Math.max(25, 45));

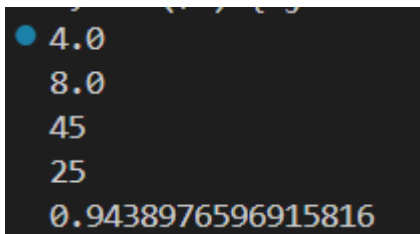
        System.out.println(Math.min(25, 45));

        System.out.println(Math.random());

    }

}

```



```

● 4.0
  8.0
  45
  25
  0.9438976596915816

```

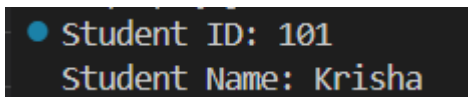
2: Classes & Objects

Exercise 9: Creating a Simple Class

Create a class named Student containing student ID and name. Create an object and display

>>

```
class Student {  
    int id;  
    String name;  
}  
  
public class StudentDetails {  
    public static void main(String[] args) {  
        Student s = new Student();  
  
        s.id = 101;  
        s.name = "Krisha";  
  
        System.out.println("Student ID: " + s.id);  
        System.out.println("Student Name: " + s.name);  
    }  
}
```



```
● Student ID: 101  
  Student Name: Krisha
```

Exercise 10: Employee Information

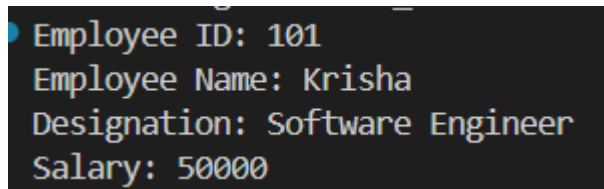
Create an Employee class with employee ID, name, designation, and salary. Create multiple objects and display their details.

>>

```
class Employee {  
    int empid;  
    String name;  
    String designation;  
    int salary;  
}  
  
public class EmployeeDetails {  
    public static void main(String[] args) {  
        Employee e = new Employee();
```

```
e.empid = 101;
e.name = "Krisha";
e.designation = "Software Engineer";
e.salary = 50000;

System.out.println("Employee ID: " + e.empid);
System.out.println("Employee Name: " + e.name);
System.out.println("Designation: " + e.designation);
System.out.println("Salary: " + e.salary);
}
}
```



A screenshot of a Java program's output. It displays four lines of text: 'Employee ID: 101', 'Employee Name: Krisha', 'Designation: Software Engineer', and 'Salary: 50000'. The text is rendered in a monospaced font with color coding (blue for keywords, red for strings, green for numbers) against a dark background.

Exercise 11: Rectangle Class

Create a Rectangle class with length and width as data members. Implement methods to calculate area and perimeter.

>>

```
class Rectangle{
    int length;
    int width;
}

public class Rect{
    public static void main(String[] args) {
        Rectangle r = new Rectangle();
        r.length = 10;
        r.width = 5;

        System.out.println("Area of Rectangle: " + (r.length * r.width));
        System.out.println("Perimeter of Rectangle: " + (2 * (r.length + r.width)));
    }
}
```



```
● Area of Rectangle: 50  
  Perimeter of Rectangle: 30
```

Exercise 12: Circle Class

Create a Circle class with radius as a data member and methods to calculate area and circumference.

>>

```
class Circle {  
    int radius;  
}  
  
public class CircleClass {  
    public static void main(String[] args) {  
        Circle c = new Circle();  
        c.radius = 7;  
  
        System.out.println("Area of Circle: " + (3.14 * c.radius * c.radius));  
        System.out.println("Circumference of Circle: " + (2 * 3.14 * c.radius));  
    }  
}
```

```
● Area of Circle: 153.86  
  Circumference of Circle: 43.96
```

Exercise 13: Bank Account

Create a BankAccount class with account number, account holder name, and balance. Implement methods for deposit, withdrawal, and balance inquiry.

>>

```
class bankaccount{  
    int acc_no;  
    String name;  
    int balance;  
}  
  
public class bnkacc {  
    public static void main(String [] args){
```

```
bankaccount b = new bankaccount();

b.acc_no = 10101;

b.name = "Krisha";

b.balance = 5000;

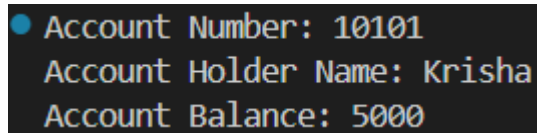

System.out.println("Account Number: " + b.acc_no);

System.out.println("Account Holder Name: " + b.name);

System.out.println("Account Balance: " + b.balance);

}

}
```

A terminal window with a dark background and light blue text. It displays three lines of output: "Account Number: 10101", "Account Holder Name: Krisha", and "Account Balance: 5000".

```
● Account Number: 10101
  Account Holder Name: Krisha
  Account Balance: 5000
```

Exercise 14: Product Management

Create a Product class with product ID, name, price, and quantity. Calculate the total cost of purchased products.

>>

```
class Product{

    int prod_id;

    String name;

    int price;

    int quantity;

}

public class productManagement {

    public static void main(String[] args) {

        Product p = new Product();

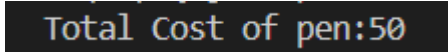

        p.prod_id = 101;

        p.name = "pen";

        p.price = 10;

        p.quantity = 5;
```

```
System.out.println("Total Cost of " + p.name + ":" + (p.price * p.quantity));  
}  
}
```



Exercise 15: Student Result

Create a Student class containing marks of five subjects. Implement methods to calculate total marks, percentage, and grade.

>>

```
class Student {  
    int m1;  
    int m2;  
    int m3;  
    int m4;  
    int m5;  
  
    int totalMarks() {  
        return m1 + m2 + m3 + m4 + m5;  
    }  
  
    double percentage() {  
        return totalMarks() / 5.0;  
    }  
  
    String grade() {  
        double p = percentage();  
  
        if (p >= 90)  
            return "A";  
        else if (p >= 80)  
            return "B";  
        else if (p >= 70)  
            return "C";  
        else if (p >= 60)
```

```

        return "D";

    else

        return "F";

    }

}

```

```

public class result {

    public static void main(String[] args) {

        Student s = new Student();


        s.m1 = 85;

        s.m2 = 90;

        s.m3 = 78;

        s.m4 = 92;

        s.m5 = 88;


        System.out.println("Total Marks: " + s.totalMarks());

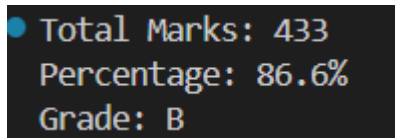
        System.out.println("Percentage: " + s.percentage() + "%");

        System.out.println("Grade: " + s.grade());

    }

}

```



```

● Total Marks: 433
  Percentage: 86.6%
  Grade: B

```

Exercise 16: Car Details

Create a Car class with brand, model, manufacturing year, and price. Display the details of multiple cars.

```
>>
```

```
//display multiple car record
```

```

class Car{

    String brand;

    String model;

    int manufacturingYear;

    int price;

```

```
}  
  
public class cardetail {  
  
    public static void main(String[] args) {  
  
        Car c1 = new Car();  
        c1.brand = "Toyota";  
        c1.model = "Camry";  
        c1.manufacturingYear = 2020;  
        c1.price = 250000;  
  
        Car c2 = new Car();  
        c2.brand = "Honda";  
        c2.model = "Civic";  
        c2.manufacturingYear = 2019;  
        c2.price = 220000;  
  
        Car c3 = new Car();  
        c3.brand = "Ford";  
        c3.model = "Mustang";  
        c3.manufacturingYear = 2021;  
        c3.price = 350000;  
  
        System.out.println("Car 1:");  
        System.out.println("Brand: " + c1.brand);  
        System.out.println("Model: " + c1.model);  
        System.out.println("Manufacturing Year: " + c1.manufacturingYear);  
        System.out.println("Price: " + c1.price);  
  
        System.out.println("\nCar 2:");  
        System.out.println("Brand: " + c2.brand);  
        System.out.println("Model: " + c2.model);  
        System.out.println("Manufacturing Year: " + c2.manufacturingYear);  
        System.out.println("Price: " + c2.price);  
  
        System.out.println("\nCar 3:");
```

```
System.out.println("Brand: " + c3.brand);

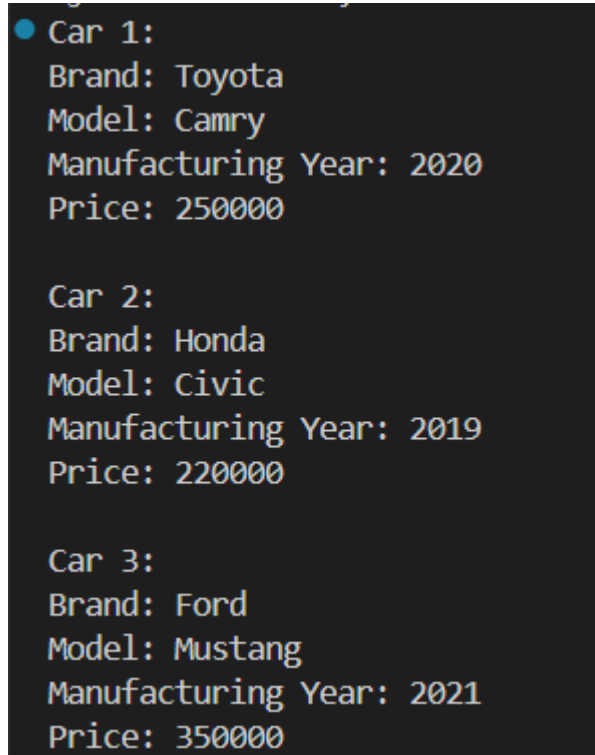
System.out.println("Model: " + c3.model);

System.out.println("Manufacturing Year: " + c3.manufacturingYear);

System.out.println("Price: " + c3.price);

}

}
```



```
● Car 1:
  Brand: Toyota
  Model: Camry
  Manufacturing Year: 2020
  Price: 250000

  Car 2:
  Brand: Honda
  Model: Civic
  Manufacturing Year: 2019
  Price: 220000

  Car 3:
  Brand: Ford
  Model: Mustang
  Manufacturing Year: 2021
  Price: 350000
```

Exercise 17: Book Information

Create a Book class with title, author, publisher, and price. Display book details using objects.

>>

```
class Book{

    String title;

    String author;

    String publisher;

    int price;

}

public class bookinfo {

    public static void main(String[] args) {

        Book b = new Book();

        b.title = "The Great Gatsby";

        b.author = "F. Scott Fitzgerald";
```

```

        b.publisher = "Scribner";

        b.price = 1500;

        System.out.println("Book 1:");

        System.out.println("Title: " + b.title);

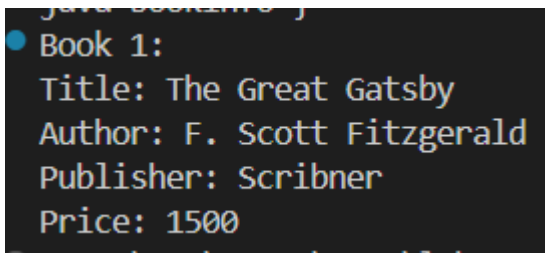
        System.out.println("Author: " + b.author);

        System.out.println("Publisher: " + b.publisher);

        System.out.println("Price: " + b.price);

    }
}

```



```

java Book1.java
Book 1:
Title: The Great Gatsby
Author: F. Scott Fitzgerald
Publisher: Scribner
Price: 1500

```

Exercise 18: Mobile Phone

Create a Mobile class with brand, model, RAM, storage, and price. Display the specifications of different mobile phones.

```
>>
```

```

class Mobile{

    String brand;

    String model;

    int RAM;

    int storage;

    int price;

}

public class MobilePhone {

    public static void main(String[] args) {

        Mobile m1 = new Mobile();

        m1.brand = "Samsung";

        m1.model = "Galaxy S21";

        m1.RAM = 8;

        m1.storage = 128;
    }
}

```

```
m1.price = 70000;
```

```
Mobile m2 = new Mobile();
```

```
m2.brand = "Apple";
```

```
m2.model = "iPhone 13";
```

```
m2.RAM = 4;
```

```
m2.storage = 256;
```

```
m2.price = 90000;
```

```
System.out.println("Mobile 1:");
```

```
System.out.println("Brand: " + m1.brand);
```

```
System.out.println("Model: " + m1.model);
```

```
System.out.println("RAM: " + m1.RAM + " GB");
```

```
System.out.println("Storage: " + m1.storage + " GB");
```

```
System.out.println("Price: " + m1.price);
```

```
System.out.println("\nMobile 2:");
```

```
System.out.println("Brand: " + m2.brand);
```

```
System.out.println("Model: " + m2.model);
```

```
System.out.println("RAM: " + m2.RAM + " GB");
```

```
System.out.println("Storage: " + m2.storage + " GB");
```

```
System.out.println("Price: " + m2.price);
```

```
}
```

```
}
```



```
● Mobile 1:  
  Brand: Samsung  
  Model: Galaxy S21  
  RAM: 8 GB  
  Storage: 128 GB  
  Price: 70000  
  
Mobile 2:  
  Brand: Apple  
  Model: iPhone 13  
  RAM: 4 GB  
  Storage: 256 GB  
  Price: 90000
```

Exercise 19: Library Management

Create a `LibraryBook` class with book ID, title, author, and availability status. Display the information of available books.

>>

```
class Library{  
    int bookId;  
    String name;  
    String author;  
    boolean isAvailable;  
}  
  
public class LibManag {  
    public static void main(String[] args) {  
        Library book1 = new Library();  
        book1.bookId = 101;  
        book1.name = "To Kill a Mockingbird";  
        book1.author = "Harper Lee";  
        book1.isAvailable = true;  
  
        Library book2 = new Library();  
        book2.bookId = 102;  
        book2.name = "1984";  
        book2.author = "George Orwell";
```

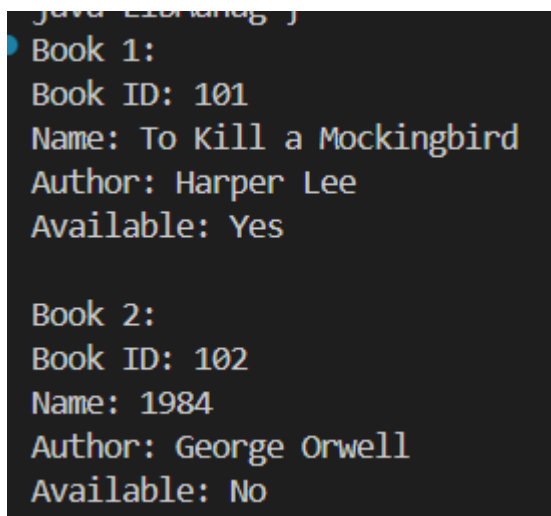
```

    book2.isAvailable = false;

    System.out.println("Book 1:");
    System.out.println("Book ID: " + book1.bookId);
    System.out.println("Name: " + book1.name);
    System.out.println("Author: " + book1.author);
    System.out.println("Available: " + (book1.isAvailable ? "Yes" : "No"));

    System.out.println("\nBook 2:");
    System.out.println("Book ID: " + book2.bookId);
    System.out.println("Name: " + book2.name);
    System.out.println("Author: " + book2.author);
    System.out.println("Available: " + (book2.isAvailable ? "Yes" : "No"));
}
}

```



```

java -E:\Programs\Java\bin\java.exe
Book 1:
Book ID: 101
Name: To Kill a Mockingbird
Author: Harper Lee
Available: Yes

Book 2:
Book ID: 102
Name: 1984
Author: George Orwell
Available: No

```

Exercise 20: Method Calling Using Objects

Create a Calculator class containing methods for addition, subtraction, multiplication, and division. Invoke these methods using an object.

>>

```

class Calculator{
    int add(int a, int b){
        return a + b;
    }
    int subtract(int a, int b){

```

```

    return a - b;
}

int multiply(int a, int b){
    return a * b;
}

double divide(int a, int b){
    return a / b;
}
}

public class MethodCalling {

    public static void main(String[] args) {

        Calculator c = new Calculator();

        System.out.println("Addition: " + c.add(20, 10));

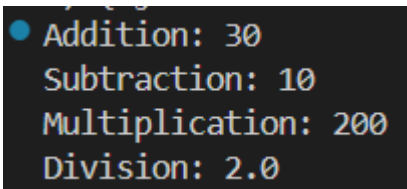
        System.out.println("Subtraction: " + c.subtract(20, 10));

        System.out.println("Multiplication: " + c.multiply(20, 10));

        System.out.println("Division: " + c.divide(20, 10));

    }
}

```



```

Addition: 30
Subtraction: 10
Multiplication: 200
Division: 2.0

```

Exercise 21: Passing Objects as Parameters

Write a Java program that passes one object to another method and displays its data.

>>

```

class Student{

    int id;

    String name;

}

public class object {

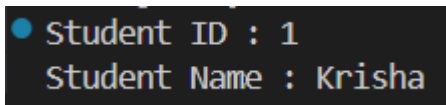
    static void display(Student s) {

        System.out.println("Student ID : " + s.id);

        System.out.println("Student Name : " + s.name);
    }
}

```

```
}  
  
public static void main(String[] args){  
  
    Student s = new Student();  
  
    s.id = 1;  
    s.name = "Krisha";  
  
    display(s);  
}  
}
```



```
● Student ID : 1  
  Student Name : Krisha
```

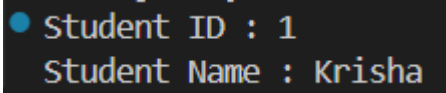
Exercise 22: Returning Objects from Methods

Write a Java program that returns an object from a method and displays its contents.

>>

```
class Student{  
    int id;  
    String name;  
}  
  
public class returnObject {  
    public static Student getStudent() {  
        Student s = new Student();  
  
        s.id = 1;  
        s.name = "Krisha";  
  
        return s;  
    }  
  
    public static void main(String[] args){  
        Student s = getStudent();  
  
        System.out.println("Student ID : " + s.id);  
        System.out.println("Student Name : " + s.name);  
    }  
}
```

```
}  
  
}
```

A terminal window with a dark background and light blue text. It displays two lines of output: "Student ID : 1" and "Student Name : Krisha".

```
Student ID : 1  
Student Name : Krisha
```

Exercise 23: Object Array

Create an array of Student objects and display the details of all students.

```
>>
```

```
class Student {  
    int id;  
    String name;  
}  
  
public class StudentObject {  
    public static void main(String[] args) {  
        Student[] students = new Student[3];  
  
        students[0] = new Student();  
        students[0].id = 1;  
        students[0].name = "Krisha";  
  
        students[1] = new Student();  
        students[1].id = 2;  
        students[1].name = "Purvi";  
  
        students[2] = new Student();  
        students[2].id = 3;  
        students[2].name = "Manashvi";  
  
        for(int i = 0; i < students.length; i++){  
            System.out.println("Student ID : " + students[i].id);  
            System.out.println("Student Name : " + students[i].name);  
            System.out.println();  
        }  
    }  
}
```

```
}  
  
● Student ID : 1  
  Student Name : Krisha  
  
  Student ID : 2  
  Student Name : Purvi  
  
  Student ID : 3  
  Student Name : Manashvi
```

Exercise 24: Comparing Objects

Create two Student objects and compare them based on their marks using a user-defined method.

>>

```
class Student{  
    String name;  
    int marks;  
}  
  
public class Comparing {  
    static void compare(Student s1, Student s2) {  
        if(s1.marks > s2.marks){  
            System.out.println(s1.name + " has scored more marks than " + s2.name);  
        } else if(s1.marks < s2.marks){  
            System.out.println(s2.name + " has scored more marks than " + s1.name);  
        } else {  
            System.out.println(s1.name + " and " + s2.name + " have scored equal marks");  
        }  
    }  
}  
  
public static void main(String[] args){  
    Student s1 = new Student();  
    s1.name = "Krisha";  
    s1.marks = 95;  
  
    Student s2 = new Student();  
    s2.name = "Purvi";  
    s2.marks = 90;
```

```
        compare(s1, s2);  
    }  
}
```

● Krishna has scored more marks than Purvi

Exercise 25: Simple Inventory System

Develop a menu-driven inventory management program using classes and objects to add, display, and search product details.

>>

```
import java.util.Scanner;
```

```
class Product {
```

```
    int id;
```

```
    String name;
```

```
    int price;
```

```
    int quantity;
```

```
}
```

```
public class inventory {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        Product[] products = new Product[10];
```

```
        int count = 0;
```

```
        int choice;
```

```
        do {
```

```
            System.out.println("1. Add Product");
```

```
            System.out.println("2. Display Products");
```

```
            System.out.println("3. Search Product");
```

```
            System.out.println("4. Exit");
```

```
            System.out.print("Enter your choice: ");
```

```
            choice = sc.nextInt();
```

```
if (choice == 1) {  
    products[count] = new Product();  
  
    System.out.print("Enter Product ID: ");  
    products[count].id = sc.nextInt();  
  
    System.out.print("Enter Product Name: ");  
    products[count].name = sc.next();  
  
    System.out.print("Enter Product Price: ");  
    products[count].price = sc.nextInt();  
  
    System.out.print("Enter Product Quantity: ");  
    products[count].quantity = sc.nextInt();  
  
    count++;  
  
    System.out.println("Product added successfully!\n");  
}  
  
else if (choice == 2) {  
    System.out.println("\nProduct List:");  
  
    for (int i = 0; i < count; i++) {  
        System.out.println("Product ID: " + products[i].id);  
        System.out.println("Product Name: " + products[i].name);  
        System.out.println("Product Price: " + products[i].price);  
        System.out.println("Product Quantity: " + products[i].quantity);  
        System.out.println();  
    }  
}  
  
else if (choice == 3) {
```



```
System.out.print("Enter Product ID to search: ");

int searchId = sc.nextInt();

for (int i = 0; i < count; i++) {
    if (products[i].id == searchId) {
        System.out.println("Product Found:");
        System.out.println("Product ID: " + products[i].id);
        System.out.println("Product Name: " + products[i].name);
        System.out.println("Product Price: " + products[i].price);
        System.out.println("Product Quantity: " + products[i].quantity);
    }
}

else if (choice == 4) {
    System.out.println("Thank you...");
}

else {
    System.out.println("Invalid choice! Please try again.\n");
}

} while (choice != 4);

sc.close();

}

}
```

```
1. Add Product
2. Display Products
3. Search Product
4. Exit
Enter your choice: 1
Enter Product ID: 101
Enter Product Name: pen
Enter Product Price: 10
Enter Product Quantity: 4
Product added successfully!

1. Add Product
2. Display Products
3. Search Product
4. Exit
Enter your choice: 2

Product List:
Product ID: 101
Product Name: pen
Product Price: 10
Product Quantity: 4

1. Add Product
2. Display Products
3. Search Product
4. Exit
Enter your choice: 2

Product List:
Product ID: 101
Product Name: pen
Product Price: 10
Product Quantity: 4

1. Add Product
2. Display Products
3. Search Product
4. Exit
Enter your choice: 4
Thank you...
PS D:\MCA\SEM-1\JAVA\lab Assignment - 1> 
```